MT1011 Engineering Materials

Phase Diagrams- Eutectic Systems

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Binary eutectic systems

Copper-Silver system

- The Ag-Cu binary system is an example of a binary eutectic phase diagram.
- 3 single-phase regions on the diagram: α, β, and liquid.

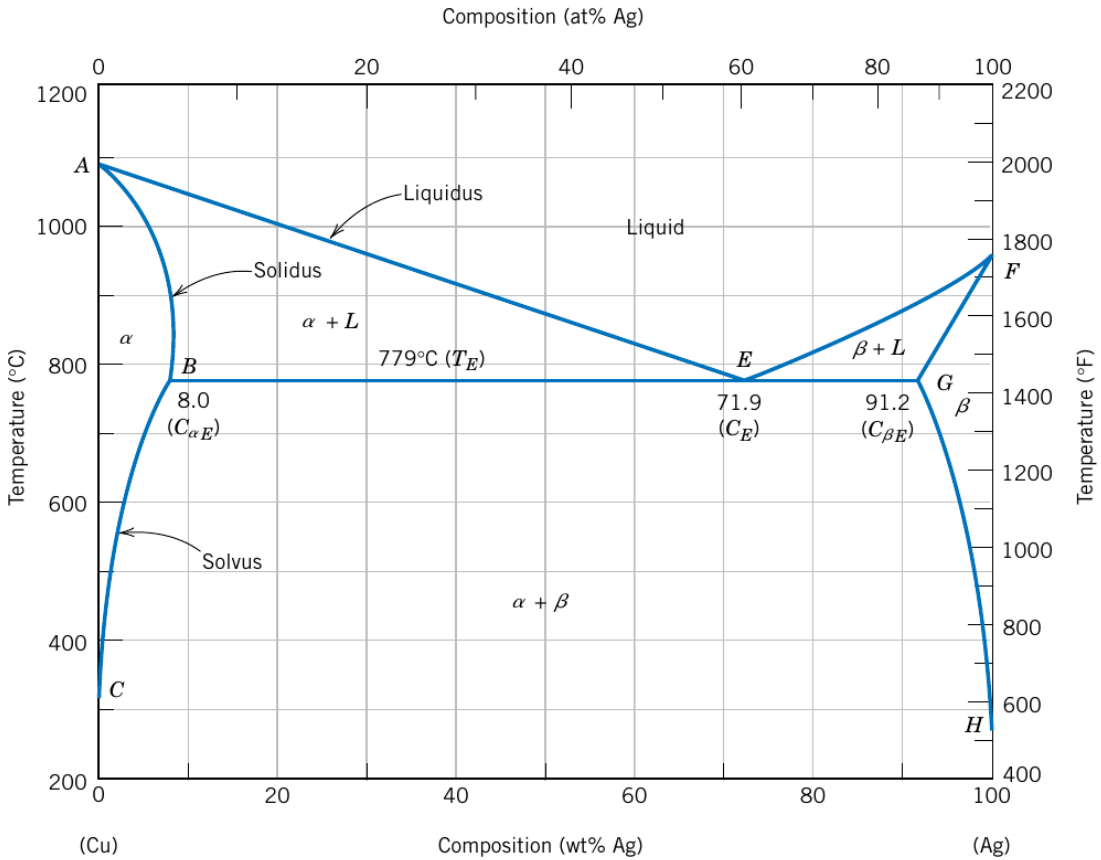


Figure: The copper–silver phase diagram.

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Phase	Type	Solute	Solvent	Crystal Structure
α	Solid solution rich in Cu	Ag	Cu	FCC
β	Solid solution rich in Ag	Cu	Ag	FCC

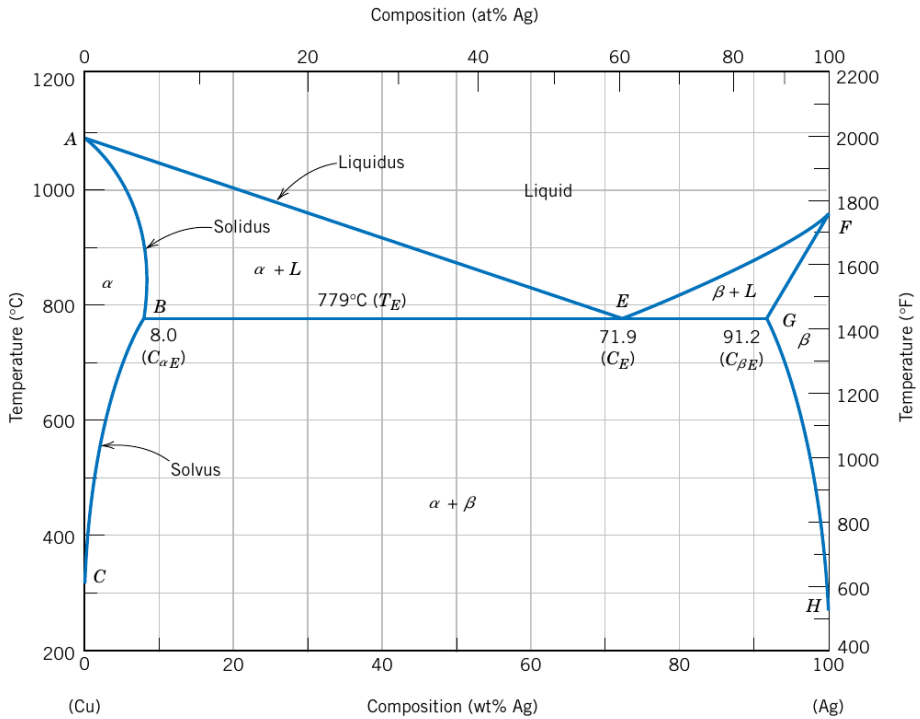
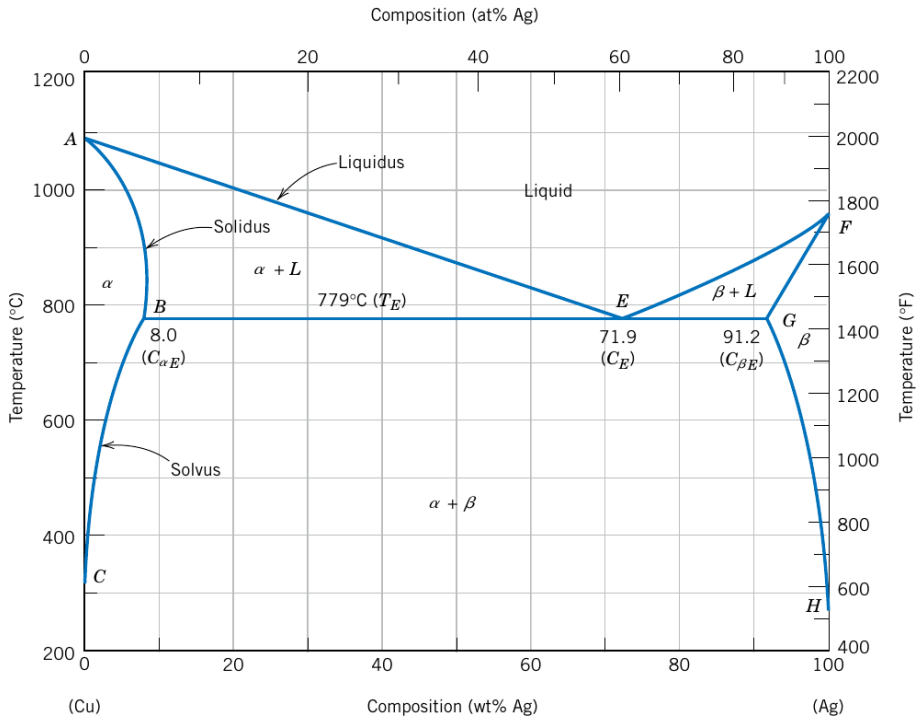


Figure: The copper–silver phase diagram.

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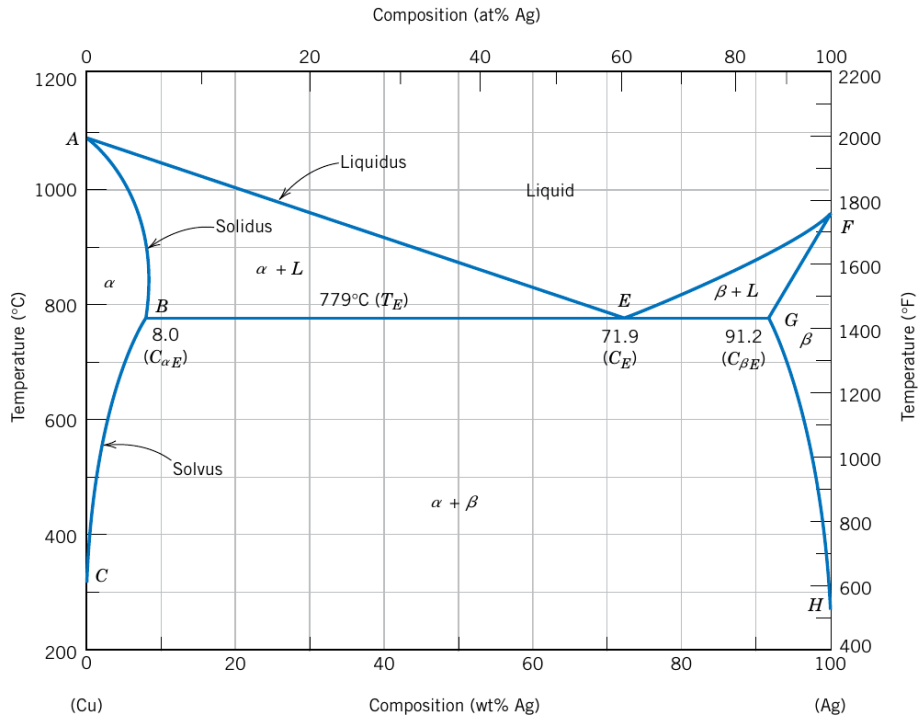
- Solubility in each of α and β solid phases is limited.
- Solubility limit for the α phase is given by the CBA boundary.
- Solubility of Ag in Cu is maximum at point B (8 wt.% Ag).
- Below 779 the solubility limit line (BC) separating the α and β phases is termed a solvus line.
- Boundary AB between α and α + Liquid fields is the solidus.



Binary eutectic systems

Copper-Silver system

- Solubility limit for the β phase is given by the . . . . . boundary.
- Solubility of Cu in Ag is maximum at . . . (91.2 wt.% Ag).
- This horizontal line BEG extends between the maximum solubility positions.
- BEG may also be considered a solidus line because it gives the lowest temperature at which a liquid phase may exist.

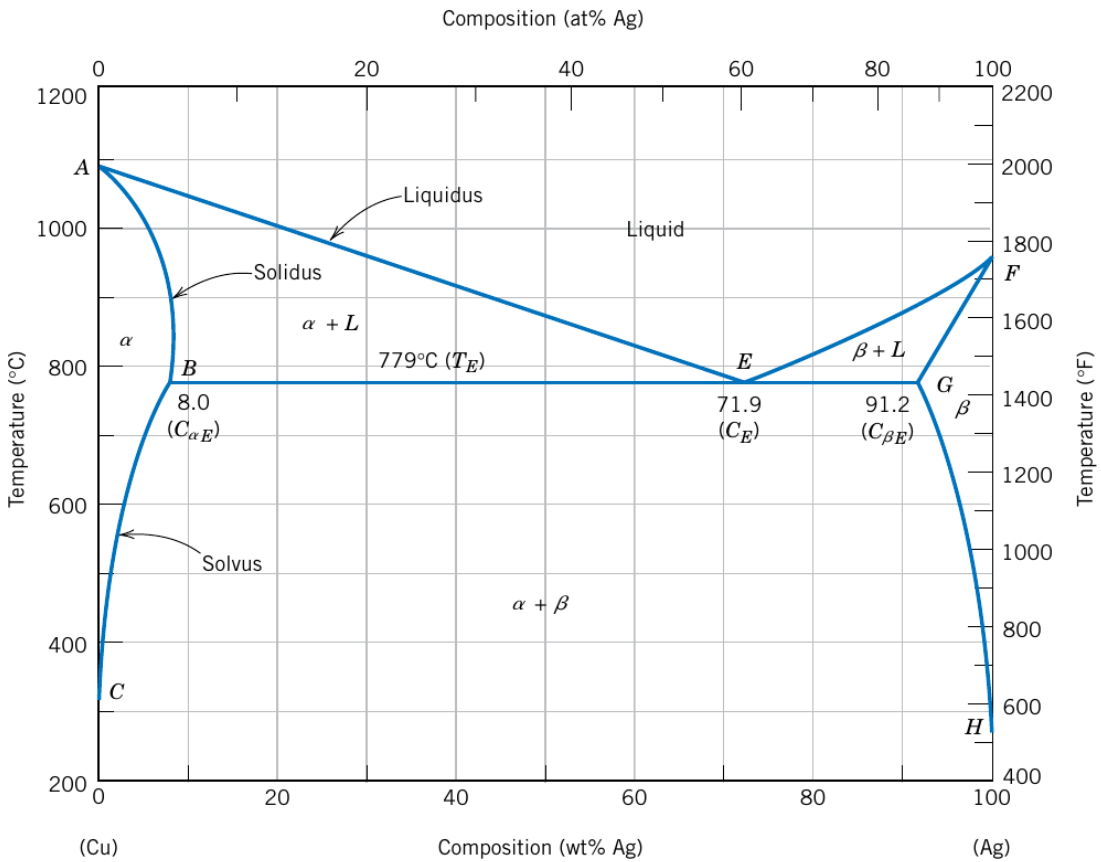


Binary eutectic systems

Copper-Silver system

Question

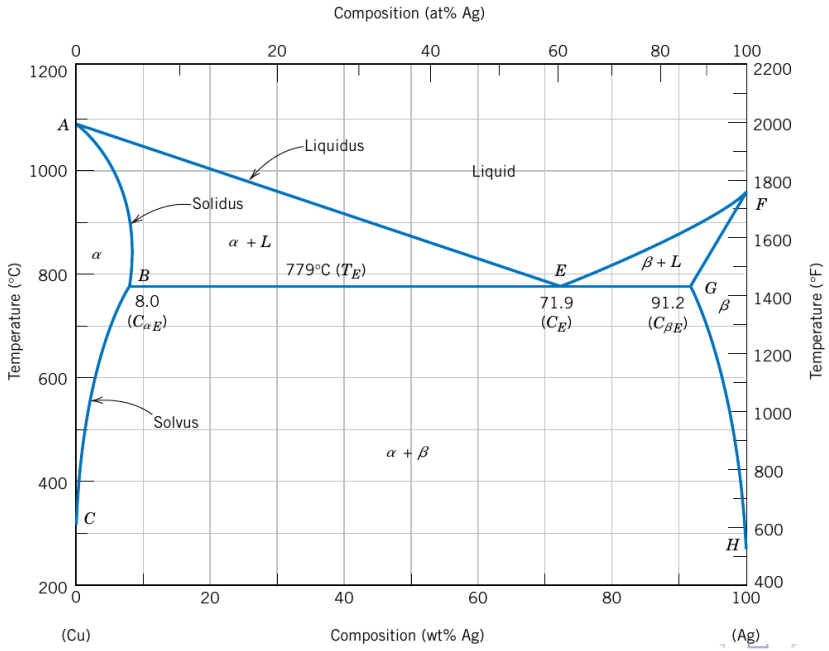
There are also three two-phase regions found for the copper–silver system. What are they?



Question

There are also three two-phase regions found for the copper–silver system. What are they? Answer:

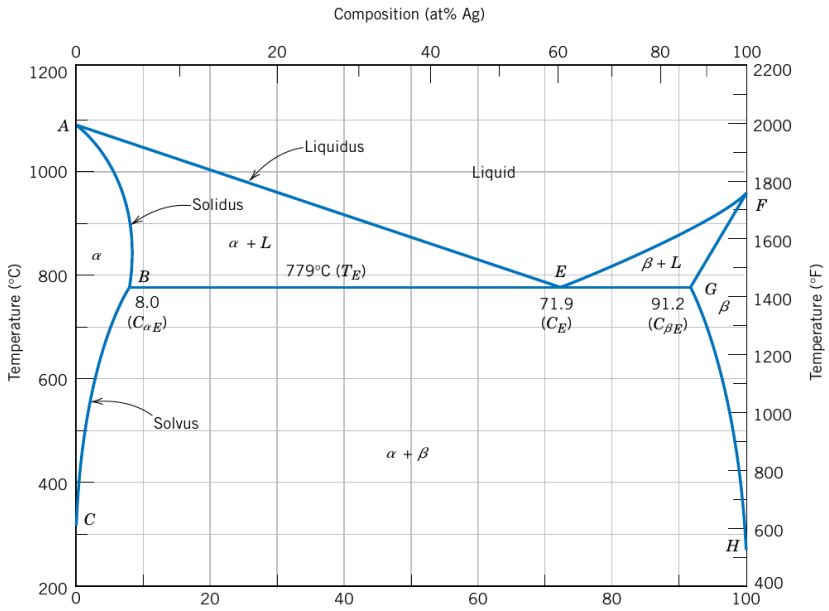
1. $\alpha + \mathrm{L}$
2. $\beta + \mathrm{L}$
3. $\alpha + \beta$



Binary eutectic systems

Copper-Silver system

- As silver is added to copper, the melting point (liquidus) decreases along the liquidus line, line AE.
- The melting temperature of Cu is lowered by silver additions.
- The same may be said for Ag.
- Introduction of Cu reduces the complete melting temperature along the other liquidus line, FE.
- The liquidus lines meet at the point E on the phase diagram.
- Point E is designated by composition C_E and temperature T_E.

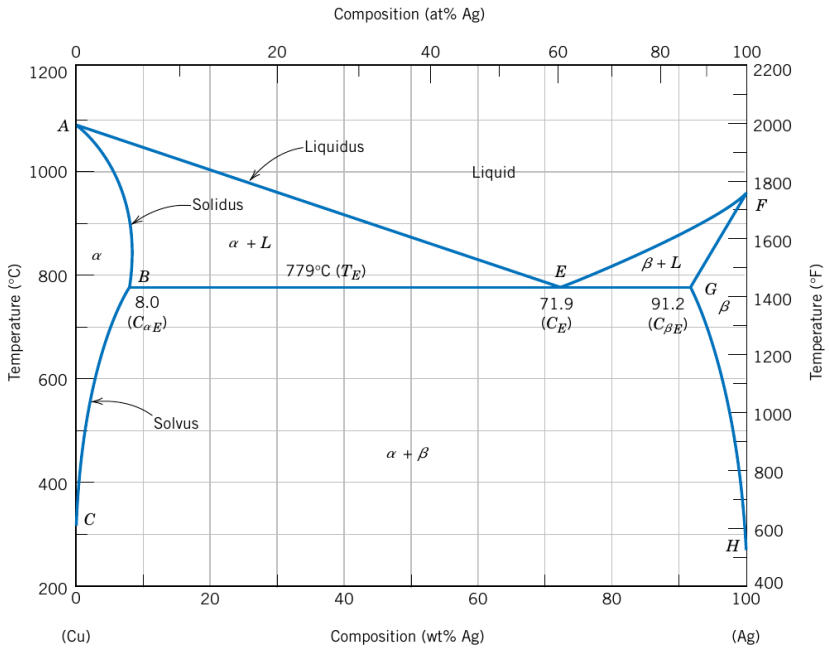


Binary eutectic systems

Copper-Silver system

$$L(C_E) \rightleftharpoons \alpha(C_{\alpha E}) + \beta(C_{\beta E})$$

- Upon cooling, a liquid phase is transformed into the two solid α and β phases at the temperature T_E.
- The opposite reaction occurs upon heating.
- This is called a eutectic reaction (eutectic = easily melted).

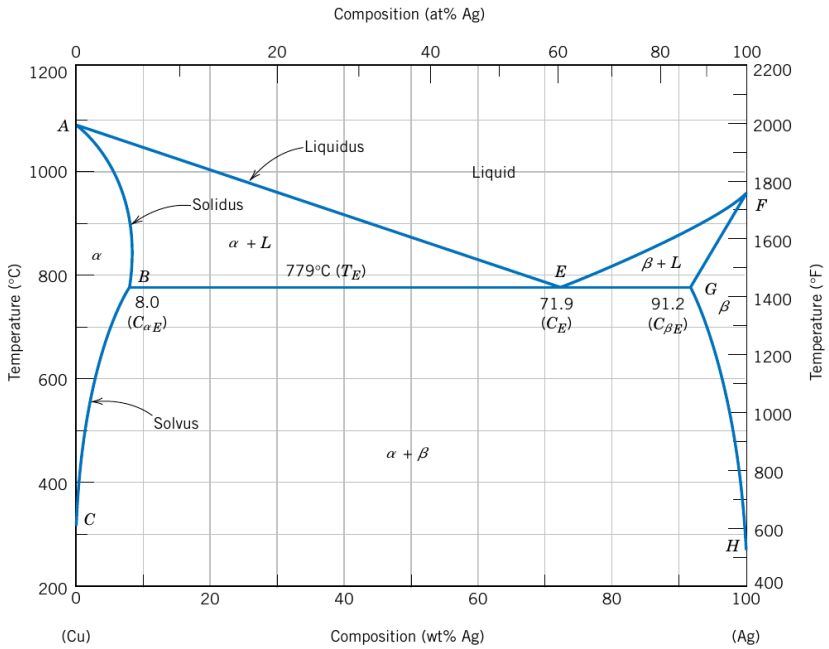


Binary eutectic systems

Copper-Silver system

$$L(C_E) \rightleftharpoons \alpha(C_{\alpha E}) + \beta(C_{\beta E})$$

- C_E and T_E represent the eutectic composition and temperature, respectively.
- $C_{\alpha E}$ and $C_{\beta E}$ are the respective compositions of the α and β phases at T_E.



First consider the following alloys in the Pb-Sn system: Pb-rich alloys containing between 0 and about 2 wt.% Sn (for the α-phase solid solution) and also between approximately 99 wt.% Sn and pure Sn.

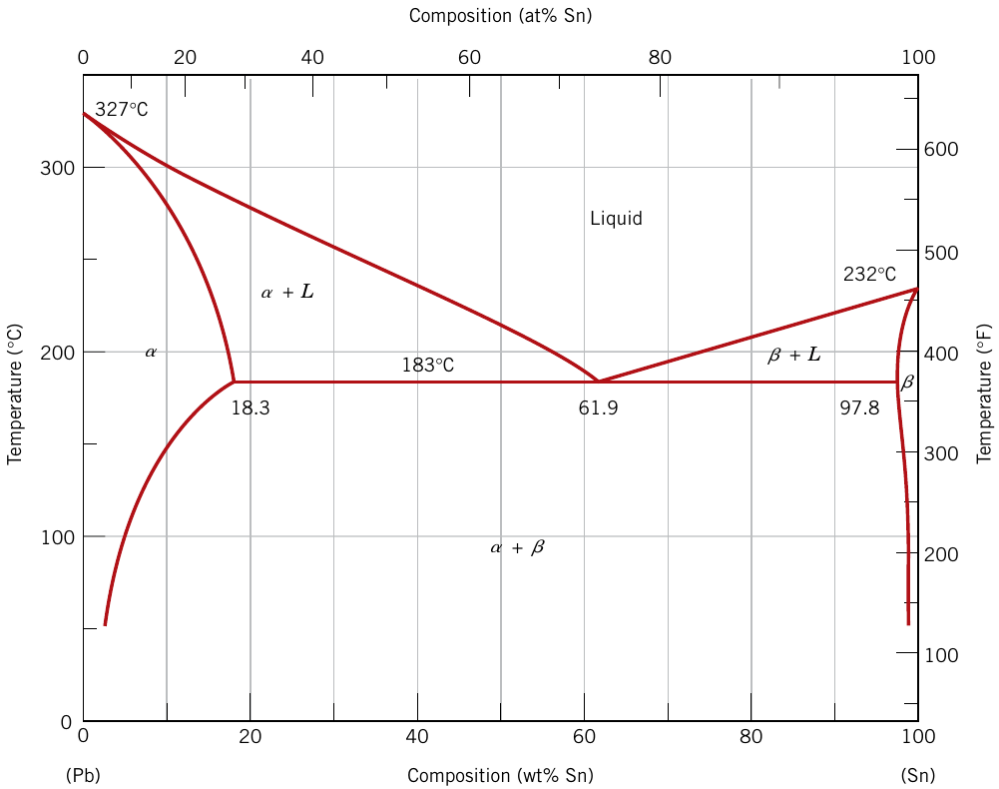
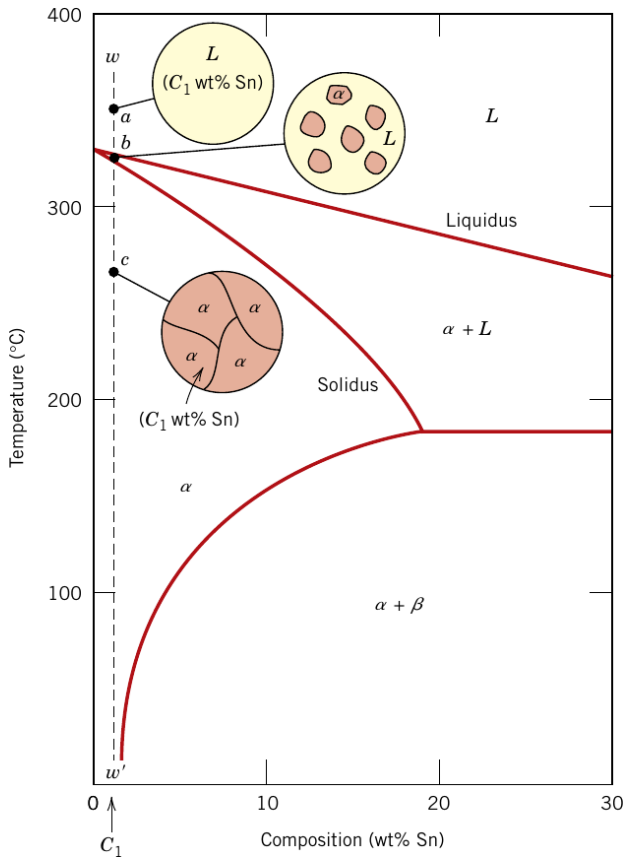


Figure: The lead–tin phase diagram.

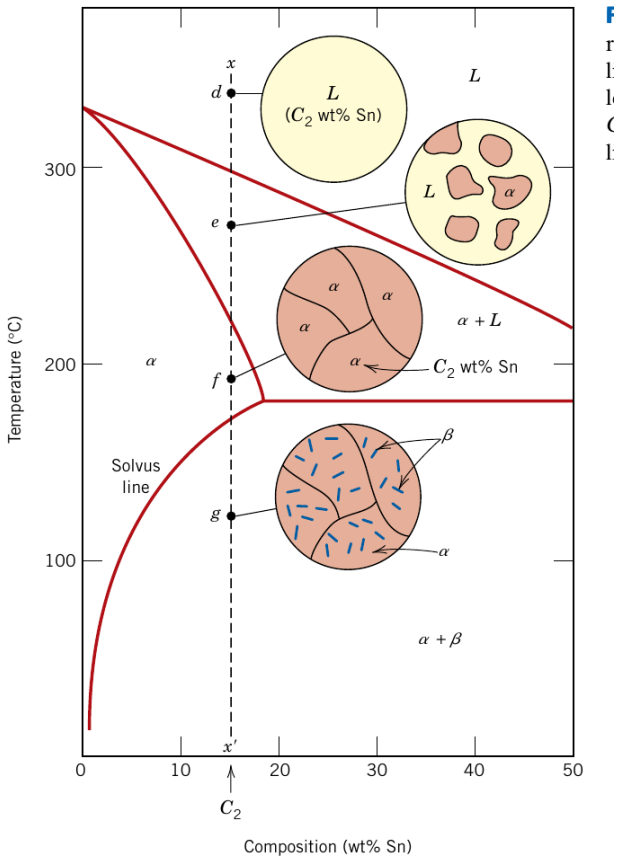
Binary eutectic systems

Development of microstructure

Describe the development of equilibrium microstructures for a lead–tin alloy of composition C_1 as it is cooled from the liquid-phase region.



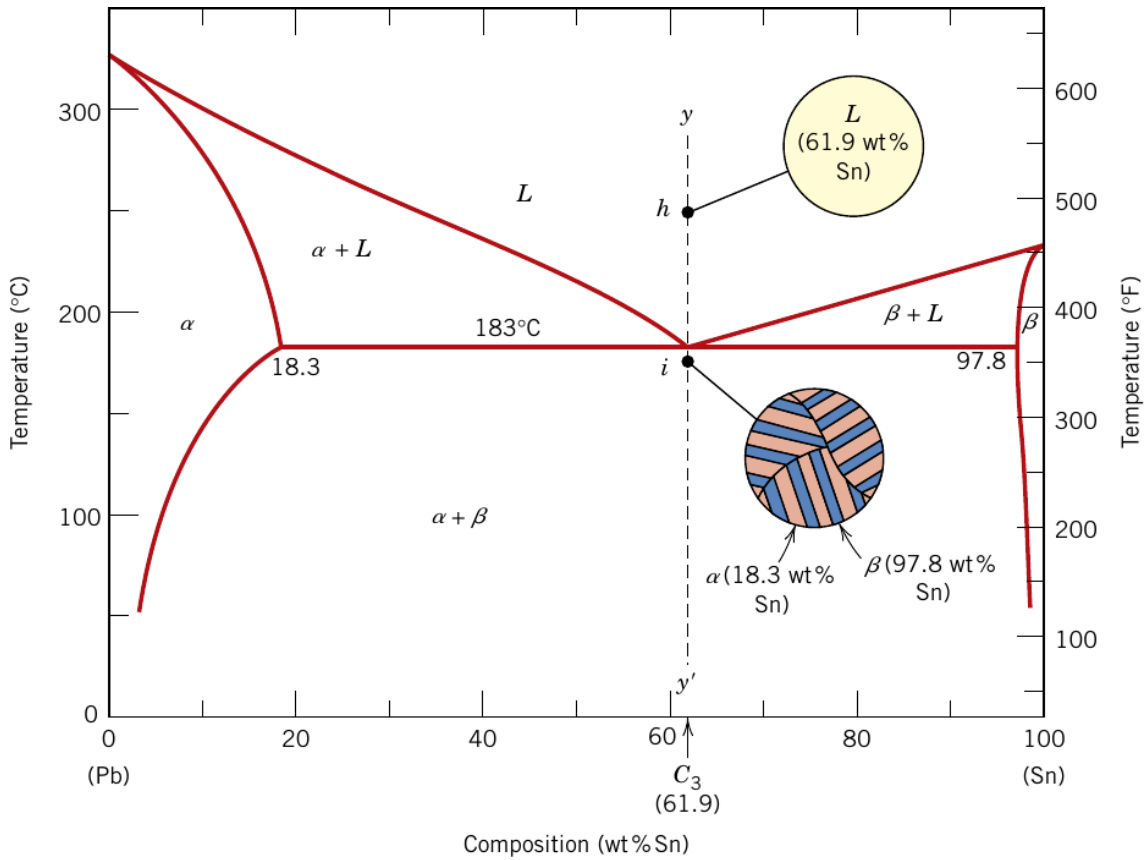
Describe the development of equilibrium microstructures for a lead–tin alloy of composition C_2 as it is cooled from the liquid-phase region.



Binary eutectic systems

Development of microstructure

Describe the development of equilibrium microstructures for a lead–tin alloy of composition C_3 as it is cooled from the liquid-phase region.



Binary eutectic systems

Development of microstructure

Describe the development of equilibrium microstructures for a lead–tin alloy of composition C_4 as it is cooled from the liquid-phase region.

